Supplementary materials

Companion to: Small vessel disease burden and outcomes of mechanical thrombectomy in ischemic stroke: a systematic review and meta-analysis

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Table S1. Search strategy in the Medline database.

Steps	Queries	Number of studies
#1	Search ((((((((((((perivascular spaces) OR white matter hyperintensities) OR white matter lacunar) OR leukoaraiosis) OR small vessel disease) OR microbleeds) OR virchow-robin spaces) OR virchow robin spaces) OR lacunes) OR lacunar) OR white matter) OR perivascular) OR virchow-robin) OR microbleed) OR small vessel	156,985
#2	Search (((((Thrombectomy) OR Endovascular) OR reperfusion) OR Recanalization) OR Aspiration) OR retriever	288,047
#1 and #2	Search ((((((((((((((perivascular spaces) OR white matter hyperintensities) OR white matter lacunar) OR leukoaraiosis) OR small vessel disease) OR microbleeds) OR virchow-robin spaces) OR virchow robin spaces) OR lacunes) OR lacunar) OR white matter) OR perivascular) OR virchow-robin) OR microbleed) OR small vessel)) AND ((((((Thrombectomy) OR Endovascular) OR reperfusion) OR Recanalization) OR Aspiration) OR retriever)	5,031

The search strategy for the Embase and the Cochrane Library database was similar to that used for the Medline database

Table S2. The definitions of small vessel disease subtypes and the confounders adjusted for the multivariate analysis in each included study.

First Author, y of publication	The definitions of HLB, CMB burden, and lacune burden	Adjusted factors in each included study
Benson 2020(1)	Fazekas scale ≥ 2	NA
Mutzenbach 2020(2)	Top 25% based on ARWMC scale*	Baseline NIHSS score, sex, recanalization outcome after MT, pre-stroke mRS, and baseline collateral grade
Mistry 2020(3)	van Swieten 3-4	Age, hypertension, baseline glucose level, baseline SBP, baseline NIHSS score, onset-to-recanalization time, bridging intravenous thrombolysis, baseline ASPECTS, and recanalization outcome after MT
Mikati 2020(4)	van Swieten 3-4	NA
Mechtouff 2020(5)	Fazekas scale >2	NA
Guo 2020(6)	van Swieten 3-4	Age, baseline SBP, history of coronary heart diseases, baseline NIHSS score, baseline ASPECTS, and baseline collateral grade
Liu 2019(7)	Fazekas scale ≥ 2	Age, sex, baseline NIHSS score, onset-to-recanalization time, baseline serum creatinine, and baseline fasting glucose, smoking history, and baseline international normalized ratio
Guo 2019(8)	van Swieten 3-4	Age, baseline SBP, baseline NIHSS score, the etiology of stroke, baseline collateral grade, recanalization outcome after MT, and history of coronary heart diseases

Boulouis 2019(9)	High WMH volume	Age, sex, baseline NIHSS score, and onset-to-groin puncture time, and recanalization outcome after MT
Arba 2019(10)	van Swieten 3-4 and Lacune presence#	Age, sex, baseline NIHSS score, and onset-to-groin puncture time
Sillanpaa 2018(11)	Lacune presence#	NA
Choi 2018(12)	CMB presence detected with an MRI scan	Age, sex, hypertension, diabetes mellitus, dyslipidemia, atrial fibrillation, smoking history, stroke or transient ischemic attack history, pre-stroke modified Rankin Scale, baseline NIHSS score, baseline infarct volume, and recanalization outcome after MT
Gilberti 2017(13)	van Swieten 2-4	NA
Atchaneeyasakul 2017(14)	High WMH volume	baseline NIHSS score, onset-to-recanalization time, histroy of congestive heart failure, and recanalization outcome after MT
Shi 2016(15)	CMB presence detected with an MRI scan	Age, baseline NIHSS score, hypertension, diabetes mellitus, atrial fibrillation, baseline glucose level, baseline SBP, and onset-to-groin puncture time>5 h
Giurgiutiu 2015(16)	High WMH volume	Age, baseline NIHSS score, baseline glucose level, baseline diffusion weighted imaging outcome, recanalization outcome after MT, bridging intravenous thrombolysis, and hypertension
Zhang 2014(17)	van Swieten 3-4	Age, baseline NIHSS score, atrial fibrillation, baseline serum creatinine, dyslipidemia, baseline glucose level, recanalization outcome

		after MT, onset-to-recanalization time, baseline collateral grade, and parenchymal hemorrhage after MT
Gratz 2014(18)	CMB presence detected with an MRI scan	NA
Soize 2013(19)	Fazekas scale ≥ 2	NA
Shi 2012(20)	Fazekas scale ≥ 2	NA

Abbreviations: NA, not available; NIHSS, National Institutes of Health Stroke Scale; MT, mechanical thrombectomy; SBP, systolic blood pressure; ASPECTS, Alberta Stroke Program Early CT Score; WMH, white matter hyperintensity; CMBs, cerebral microbleeds.

*Leukoaraiosis severity assessed with Top 25% based on ARWMC scale (a semi-quantitative scale similar with VSS or FS)(2).

Lacune burden was defined as the presence of multiple lacunes (≥ 2) detected with a CT or MRI scan (10).

Table S3. Quality assessment of the included studies*

Reference#	Is the exposed cohort representative?	Selection of the non-exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of important factors†	Assessment of outcome	Follow up period	Adequacy of follow up of cohorts	Total quality scores
Benson 2020(1)	☆	☆	☆	☆	☆	☆	☆	☆	8
Mutzenbach 2020(2)	☆	☆	☆	☆	☆	☆	☆	☆	8
Mistry 2020(3)	☆	☆	☆	☆	☆	☆	☆	☆	8
Mikati 2020(4)	☆	☆	☆	☆	—	☆	☆	—	6
Mechtouff 2020(5)	☆	☆	☆	☆	☆	☆	—	—	6
Guo 2020(6)	☆	—	☆	☆	☆ ☆	☆	—	—	6
Liu 2019(7)	☆	☆	☆	☆	☆ ☆	☆	—	—	7
Guo 2019(8)	☆	☆	☆	☆	☆ ☆	☆	—	—	7
Boulouis 2019(9)	☆	—	☆	☆	☆ ☆	☆	☆	☆	8
Arba 2019(10)	☆	☆	☆	☆	—	☆	—	—	5
Sillanpaa 2018(11)	☆	☆	☆	☆	—	☆	—	—	5
Choi 2018(12)	☆	☆	☆	☆	☆ ☆	☆	☆	☆	9
Gilberti 2017(13)	☆	☆	☆	☆	—	☆	—	—	5
Atchaneeyasakul(14)	☆	☆	☆	☆	—	☆	—	—	5
Shi 2016(15)	☆	☆	☆	☆	☆ ☆	☆	☆	☆	9
Giurgiutiu 2015(16)	☆	—	☆	☆	☆ ☆	☆	—	—	6
Zhang 2014(17)	☆	☆	☆	☆	—	☆	—	—	5
Gratz 2014(18)	☆	☆	☆	☆	—	☆	—	—	5
Soize 2013(19)	☆	☆	☆	☆	—	☆	—	—	5
Shi 2012(20)	☆	☆	☆	☆	☆	☆	☆	☆	8

*Newcastle-Ottawa Scale was used to assess the study quality in this meta-analysis(21). The full score was 9 stars, and the high-quality study was defined as a study with 8 awarded stars. †A maximum of two stars could be awarded for this item. One star with adjustment for age, two stars if there were additional population demographics or comorbidities.

Table S4. Meta-analysis of the association between SVD subtypes and outcomes of MT.

Outcomes	Number of studies	Sample size	OR	95% CI	p-values	I^2
HLB (VSS 3-4 or FS ≥ 2) and outcomes of MT						
90-day poor functional outcome*(3-5,7,8,10,17,19,1,2)	10	2,004	2.70	2.01–3.63	< 0.001	18.90%
90-day mortality(3,7,8,19)	4	818	1.59	0.99-2.55	0.056	0.00%
In-hospital death*(2,20)	2	314	4.06	1.48–11.13	0.006	54.00%
Post-MT ICH(5,20,1,2)	4	697	1.07	0.53–2.14	0.858	67.30%
Post-MT sICH(3,7,8,19,2)	5	1,005	1.81	1.00–3.29	0.051	0.00%
Successful recanalization (3,5)	2	598	1.16	0.73-1.86	0.528	0.00%
Futile recanalization*(8,13,1)	3	493	5.00	2.86–8.73	< 0.001	0.00%
Early neurological improvement(3,6)	2	662	0.58	0.19-1.72	0.324	77.80%
Early neurological deterioration*(6)	1	273	2.65	1.09-6.45		
HLB (volumetric software) and outcomes of MT						
90-day poor functional outcome#(9,14,16)	3	625	1.04	1.02-1.07	0.001	0.00%
90-day mortality(9,14)	2	552	1.13	0.79-1.62	0.515	45.4%
Post-MT ICH(14)	1	56	1.06	0.60-1.87		
Post-MT sICH(9)	1	496	0.99	0.93-1.04		
Successful recanalization(14)	1	56	1.30	0.70-2.38		
CMBs and outcomes of MT						
90-day poor functional outcome&(12,18)	2	1,924	1.84	1.17-2.90	0.008	0.00%
90-day mortality(18)	1	392	2.28	1.04-4.99		
In-hospital death(15)	1	206	0.57	0.17-1.84		
Post-MT ICH(15,18)	2	598	0.60	0.27-1.32	0.204	0.00%
Post-MT sICH(12,18)	2	1,924	1.42	0.19-10.55	0.733	69.20%
Lacunes and outcome of MT						
90-day poor functional outcome(10,11)	2	243	2.23	0.95-5.22	0.066	0.00%

Abbreviations: OR, odds ratio; CI, confidence interval; MT, mechanical thrombectomy; HLB, high leukoaraiosis burden; ICH, intracranial hemorrhage; sICH, symptomatic intracranial hemorrhage; CMBs, cerebral microbleeds.

*$p < 0.007$ was considered statistically significant based on Bonferroni corrections.

#p < 0.010 was considered statistically significant based on Bonferroni corrections.

&p < 0.010 was considered statistically significant based on Bonferroni corrections.

Table S5. Subgroup analyses regarding the association between HLB (assessed with VSS 3-4 or FS ≥ 2) and poor functional outcome at 90 days.

Variables	**Number of studies**	**Sample size**	**OR**	**95% CI**	**p-values**	I^2	**PI**
Assessment scales							0.237
VSS 3-4	6	1381	3.22	2.09-4.97	p < 0.001	31.40%	
FS ≥ 2	4	623	2.11	1.44-3.10	p < 0.001	0.00%	
Adjusted for confounders							0.328
Yes	6	1334	3.14	2.05-4.82	p < 0.001	28.20%	
No	4	670	2.18	1.49-3.18	p < 0.001	0.00%	
Adjusted for age							0.418
Yes	5	1041	3.18	1.90-5.33	p < 0.001	46.20%	
No	5	963	2.28	1.60-3.25	p < 0.001	0.00%	
Adjusted for baseline NIHSS score							0.328
Yes	6	1334	3.14	2.05-4.82	p < 0.001	28.20%	
No	4	670	2.18	1.49-3.18	p < 0.001	0.00%	
Adjusted for baseline collateral grade							0.047
Yes	2	380	5.25	2.77-9.93	p < 0.001	0.00%	
No	8	1624	2.27	1.70-3.02	p < 0.001	0.00%	
Adjusted for stroke onset to recanalization time							0.184
Yes	3	615	3.77	2.24-6.34	p < 0.001	0.00%	
No	7	1389	2.34	1.70-3.23	p < 0.001	9.50%	
Adjusted for recanalization outcome after MT							0.240
Yes	3	811	3.59	2.17-5.95	p < 0.001	0.10%	
No	7	1193	2.38	1.70-3.33	p < 0.001	14.10%	
Recanalization rate							0.053
>50%	7	437	2.27	1.69-3.04	p < 0.001	0.00%	
≤50%	2	1438	3.68	1.78-7.60	p < 0.001	0.00%	

Abbreviations: OR, odds ratio; CI, confidence interval; MT, mechanical thrombectomy; PI, P interaction.; HLB, high leukoaraiosis burden; NIHSS, National Institutes of Health Stroke Scale.

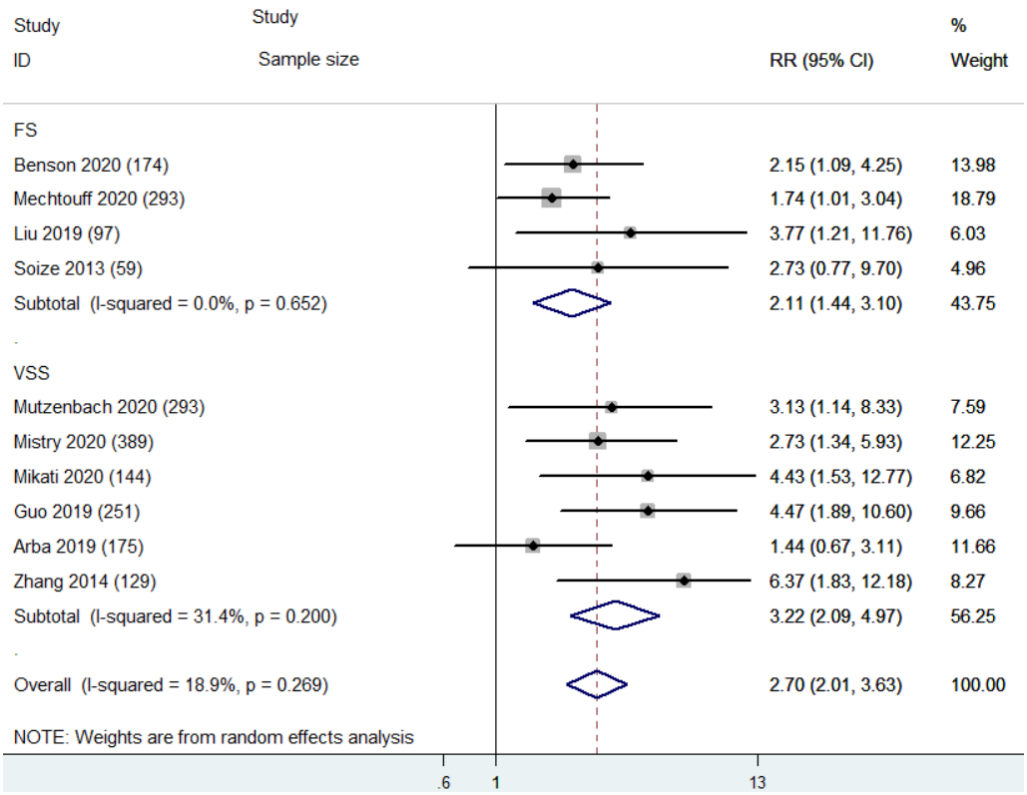

Figure S1. Subgroup analyses for the association between high leukoaraiosis burden and poor functional outcome at 90 day after mechanical thrombectomy based on variable leukoaraiosis assessment strategies: van Swieten scale 3-4 and Fazekas scale ≥ 2.

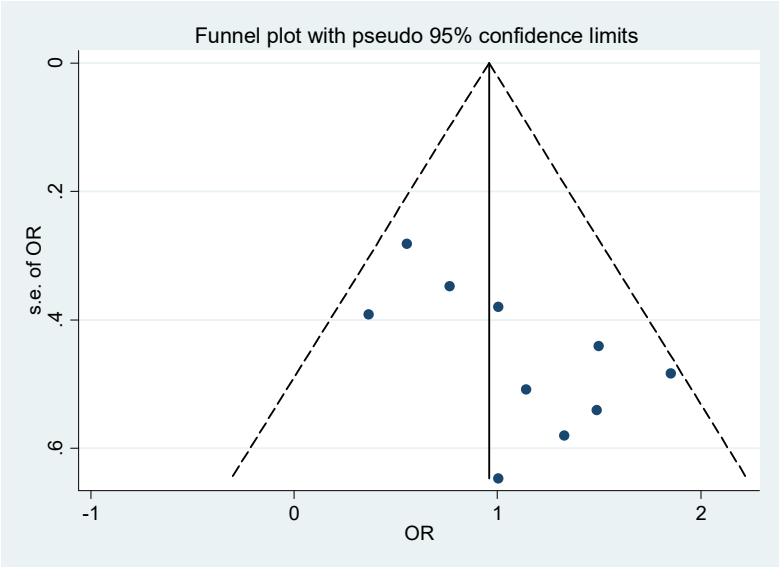

Figure S2. Funnel plot of pooled estimate for the association between high leukoaraiosis burden and poor functional outcome at 90 day after mechanical thrombectomy.

Table S6. Egger's tests and "trim and fill" analyses for publication bias.

Variables	Egger's tests		Adjusted pooled ORs and 95% CIs from "trim and fill" analyses
	p-values	95% CIs	
Association between HLB and poor functional outcome at 90 days(3-5,7,8,10,17,19,1,2)	0.037	0.23-5.50	2.11(1.51-2.94)

Abbreviations: CI, confidence interval; OR, odds ratio; HLB, high leukoaraiosis burden.

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